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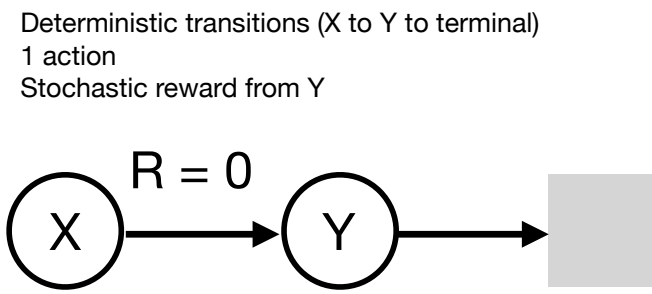
CMPUT 365: Introduction to Reinforcement Learning

Worksheet 6: Temporal Difference Learning Methods for Prediction

Question 1. Assume the agent interacts with a simple two-state MDP shown below. Every episode begins in state X, and ends when the agent transitions from state Y to the terminal state (denoted by gray box). Let's denote the set of states as $\mathcal{S} = \{X, Y, \top\}$. There is only one possible action in each state, so there is only one possible policy in this MDP. Let's denote the set of actions $\mathcal{A} = \{A\}$. In state Y the agent terminates when it takes action A and sometimes gets a reward of +1000, and sometimes gets a reward of -1000: the reward on this last transition is stochastic. Let $\gamma = 1.0$.



$$P(R = r|Y) = \begin{cases} 0.5 & \text{if } r = -1000 \\ 0.5 & \text{if } r = +1000 \end{cases}$$

(a) Write down $\pi(a|s)\ \forall\ s \in \mathcal{S}, a \in \mathcal{A}$.

(b) Write down all the possible trajectories (sequence of states, actions, and rewards) in this MDP that start from state X.

(c) What is the value of policy π (i.e. what is $v_\pi(X), v_\pi(Y)$)?

(d) Assume our estimate is equal to the value of π. That is $V(s) = v_\pi(s)\ \forall\ s \in \mathcal{S}$. Now compute the TD-error $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$ for the transition from state Y to the terminal state, assuming $R_{t+1} = +1000$. Why is the TD-error not zero if we start with $V(Y) = v_\pi(Y)$?

(e) Based on your answer to (d), what does this mean for the TD-update, for constant $\alpha = 0.1$? Will $V(Y) = v_\pi(Y) = 0$ after we update the value? Recall the TD-update is $V(S_t) \leftarrow V(S_t) + \alpha\delta_t$. What does this tell us about the updates TD(0) would make on this MDP?

(f) What is the expected TD-update, from state Y for the given V?

(g) Assume still that $V = V_\pi = 0$. What is the expectation and the variance of the TD update from state X? What is the expectation and the variance of the Monte-carlo update from state X?

(a) $\pi(A|X) = 1$ and $\pi(A|Y) = 1$.

(b) $(S_0 = X, A_0 = A, R_1 = 0, S_1 = Y, A_1 = A, R_2 = 1000, S_2 = \top)$ and $(S_0 = X, A_0 = A, R_1 = 0, S_1 = Y, A_1 = A, R_2 = -1000, S_2 = \top)$.

(c) We know that $v_\pi(S) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)[r + \gamma v_\pi(s')]$ and that $v_\pi(\top) \doteq 0$. Using this $v_\pi(Y) = 1 \cdot [0.5(1000 + \gamma v_\pi(\top)) + 0.5(-1000 + \gamma v_\pi(\top))] = 0$ and $v_\pi(X) = 1 \cdot 1 \cdot [0 + \gamma v_\pi(Y)] = 0$.

(d) $\delta_t = R_{t+1} + \gamma v_\pi(S_{t+1}) - v_\pi(S_t) = 1000 + \gamma v_\pi(\top) - v_\pi(Y) = 1000$. The TD error is zero in expectation for the true value function, i.e., $\mathbb{E}[\delta_t | S_t] = 0$ for v_π as we later show. Its value doesn't have to be zero.

(e) Based on the above sub–question, if we make the TD update for state Y using $\alpha = 0.1$, we would obtain: $V(Y) = V(Y) + 0.1 \times 1000 = 100$. If we continue to make such updates, we can see that $V(Y)$ will oscillate without ever going to zero.

(f) $\mathbb{E}[\delta_t | S_t] = \sum_r p(r|Y)(r + \gamma v_\pi(\top) - v_\pi(Y)) = 0.5(1000 + \gamma v_\pi(\top) - v_\pi(Y)) + 0.5(-1000 + \gamma v_\pi(\top) - v_\pi(Y)) = 0$.

(g) TD update: The expecation of the TD update is:

$$\mathbb{E}[\delta | X] = \mathbb{E}[0 + \gamma v_\pi(Y) - v_\pi(X) | X] = 0,$$

since each $v_\pi(X) = v_\pi(Y) = 0$. The variance of the TD update is

$$\begin{aligned}\mathbb{V}[\delta | X] &= \mathbb{E}[\delta^2 | X] - \mathbb{E}[\delta | X]^2 \\ &= \mathbb{E}[(0 + \gamma v_\pi(Y) - v_\pi(X))^2 | X] \\ &= 0.\end{aligned}$$

MC Update: The expectation of the MC update is:

$$\mathbb{E}[G - v_\pi(X) | X] = 0.5 \times 1000 + 0.5 \times (-1000) = 0.$$

Similarly, the variance is:

$$\begin{aligned}\mathbb{V}[G - v_\pi(X) | X] &= \mathbb{E}[(G - v_\pi(X))^2 | X] - \mathbb{E}[G - v_\pi(X) | X]^2 \\ &= \mathbb{E}[(G - v_\pi(X))^2 | X] \\ &= 0.5 \times 1000^2 + 0.5 \times (-1000)^2 \\ &= 1000^2.\end{aligned}$$

Question 2. Suppose that in an environment, state transitions are deterministic and that reward is bounded so that $R_{min} = 0$ and $R_{max} = 1$, with $\mathbb{E}[R_t] = 0.5$. Find the maximum and minimum possible TD error $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$, where $\gamma = 0.9$ and $V = v_\pi$ with deterministic policy π.

We know that for the exact value function v_π,

$v_\pi(S_t) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t] = \mathbb{E}[R_{t+1} | S_t] + \gamma v_\pi(S_{t+1})$.

Therefore we can write

$$\begin{aligned}\delta_t &= R_{t+1} + \gamma v_\pi(S_{t+1}) - v_\pi(S_t) \\ &= R_{t+1} + \gamma v_\pi(S_{t+1}) - \mathbb{E}[R_{t+1} | S_t] - \gamma v_\pi(S_{t+1}) \\ &= R_{t+1} - \mathbb{E}[R_{t+1} | S_t] \\ &= R_{t+1} - 0.5.\end{aligned}$$

Therefore, the maximum TD error is $\delta = R_{\max} - 0.5 = 0.5$ and the minimum TD error is $\delta = R_{\min} - 0.5 = -0.5$.

Question 3. Modify the Tabular TD(0) algorithm for estimating v_π, to estimate q_π.

Tabular TD(0) for estimating v_π

Input: the policy π to be evaluated
Algorithm parameter: step size $\alpha \in (0, 1]$
Initialize $V(s)$, for all $s \in \mathcal{S}^+$, arbitrarily except that $V(terminal) = 0$

Loop for each episode:
  Initialize S
  Loop for each step of episode:
    $A \leftarrow$ action given by π for S
    Take action A, observe R, S'
    $V(S) \leftarrow V(S) + \alpha\big[R + \gamma V(S') - V(S)\big]$
    $S \leftarrow S'$
  until S is terminal

Tabular TD(0) for estimating q_π

Input: the policy π to be evaluated
Algorithm parameter: step size $\alpha \in (0, 1]$
Initialize $Q(s, a)$, for all $s \in \mathcal{S}^+$, $a \in \mathcal{A}(s)$, arbitrarily except that $Q(terminal, \cdot) = 0$

Loop for each episode:
  Initialize S
  Choose action A given by π for S
  Loop for each step of episode
    Take action A, observe R, S'
    Choose action A' given by π for S'
    $Q(S, A) \leftarrow Q(S, A) + \alpha[R + \gamma Q(S', A') - Q(S, A)]$
    $S \leftarrow S'; A \leftarrow A';$
  until S is terminal.

Question 4. **[Challenge]** Exercise 6.1 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 5. Exercise 6.2 from Sutton & Barto (2018).

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Question 6. Exercise 6.3 from Sutton & Barto (2018).

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Question 7. Exercise 6.4 from Sutton & Barto (2018).

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Question 8. Exercise 6.5 from Sutton & Barto (2018).

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Question 9. Exercise 6.6 from Sutton & Barto (2018).

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Question 10. Exercise 6.7 from Sutton & Barto (2018).

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